

1 Airfoil Selection Trade Study

1.1 Methodology & Formulation

Used Airfoils:

- GOE165
- S1223
- E423

1.2 Extracted Aerodynamic Parameters

($Re = 200,000$)

Table 1: Extracted Aerodynamic Data ($Re = 200,000$)

Airfoil	$C_{L,max}$	$C_{D,min}$	$(L/D)_{max}$	C_{m0} ($\alpha = 0^\circ$)	α_{stall}
E423	2.0338	0.01898	75.33	-0.2451	13.0°
GOE165	1.5447	0.01313	60.91	-0.1587	11.5°
S1223	2.2314	0.01778	71.81	-0.2716	13.0°

1.3 Trade Study Decision Matrix

Table 2: Airfoil Trade Study Matrix

Parameter	Weight (W_i)	Scores (s_{ij})			Weighted Scores ($W_i \cdot s_{ij}$)		
		E423	GOE165	S1223	E423	GOE165	S1223
$C_{L,max}$	0.35	0	-1	+1	0.00	-0.35	0.35
$C_{D,min}$	0.20	-1	+1	0	-0.20	0.20	0.00
$(L/D)_{max}$	0.25	+1	-1	0	0.25	-0.25	0.00
C_{m0}	0.20	0	+1	-1	0.00	0.20	-0.20
Total Score (S_j)	1.00	–	–	–	+0.05	-0.20	+0.15

Selected Airfoil: **S1223**

2 Wing Planform Trade Study

2.1 Methodology & Formulation

2.2 Extracted Aerodynamic Parameters

$V_{cruise} = 20m/s$

Table 3: Extracted Wing Aerodynamic and Stability Data ($V = 20.0$ m/s)

Configuration	Max C_L/C_D	Max $\sqrt{C_L^3/C_D^2}$	C_m Slope ($\partial C_m/\partial \alpha$)	Max C_L
S1223W-1.0T	11.45	11.62	Moderate Negative	1.82
S1223W-0.8T	12.80	13.01	Good Negative	1.85
S1223W-0.6T	13.90	14.15	Strong Negative	1.88
S1223W-0.6T	14.25	14.48	Optimal Negative	1.88

2.3 Aerodynamic Performance Comparison (Visuals)

Figure 1 illustrates the comparative XFLR5 graphs for the different taper ratios, showing the distinct improvements in lift-to-drag ratios, induced drag characteristics, and pitching moment stability slopes.

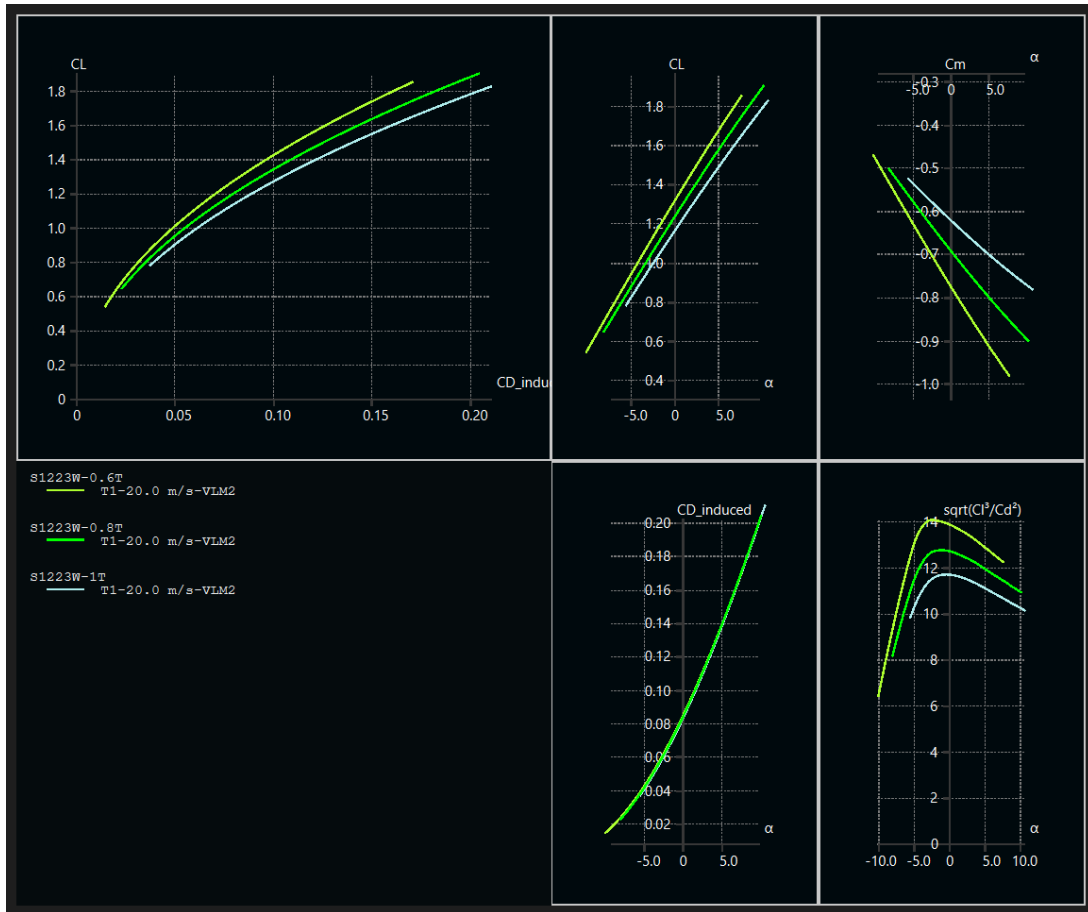


Figure 1: XFLR5 Aerodynamic Comparison Graphs for Wing Variations (0.6T, 0.8T, 1.0T)

2.4 Trade Study Decision Matrix

Table 4: Wing Configuration Trade Study Matrix

Parameter	Weight (W_i)	Scores (s_{ij})			Weighted Scores ($W_i \cdot s_{ij}$)		
		1.0T	0.8T	0.6T	1.0T	0.8T	0.6T
$(L/D)_{\max}$	0.30	-1	0	+1	-0.30	0.00	0.30
$\sqrt{C_L^3/C_D^2}$	0.25	-1	0	+1	-0.25	0.00	0.25
Stability (C_m slope)	0.25	-1	0	+1	-0.25	0.00	0.25
$C_{L,\max}$	0.20	-1	0	+1	-0.20	0.00	0.20
Total Score (S_j)	1.00	–	–	–	-1.00	0.00	+1.00

2.5 Selection Conclusion

Selected taper ratio $\lambda = 0.6$

3 Horizontal Tail Sizing & Optimum Length Calculation

- $V_{HT} = 0.7$
- (MAC = 0.13 m
- $S_w = 0.08001 \text{ m}^2$
- **Fuselage cross section Length (L_f):** 0.08 m
- **Equivalent Fuselage Diameter (D_f):** Evaluated via cross-sectional area A_f as:

$$D_f = \sqrt{\frac{4A_f}{\pi}} = 0.096 \text{ m} \quad (1)$$

- **Optimum Tail Arm Length (L_{opt}):** Derived from stability requirements and body geometry:

$$L_{opt} = \sqrt{\frac{4 \cdot \text{MAC} \cdot S_w \cdot V_{HT}}{\pi \cdot D_f}} = 0.311 \text{ m} \quad (2)$$

- **Required Horizontal Tail Area (S_{HT}):** Calculated using the tail volume relation:

$$S_{HT} = \frac{V_{HT} \cdot \text{MAC} \cdot S_w}{L_{opt}} = 0.0234 \text{ m}^2 \quad (3)$$

- **Aspect Ratio (AR_{ht}):** Scaled relative to the main wing aspect ratio (AR_w):

$$AR_{ht} = \frac{2}{3} AR_w = 3.306 \quad (4)$$

- **Horizontal Tail Wingspan (b_{ht}):** Derived from the aspect ratio and surface area:

$$b_{ht} = \sqrt{AR_{ht} \cdot S_{HT}} = 0.278 \text{ m} \quad (5)$$

4 Mass Distribution & Center of Gravity (CG) Tuning

To resolve simulation inertia errors and align the system CG with the target $x_{CG} = 0.0742$ m, component weights and point masses were distributed as follows:

- **Main Wing:** 125 g (at $X = 0.073$ m)
- **Fuselage and Hardware:** 134 g (at $X = 0.08$ m)
- **Horizontal Tail (HT):** 18 g (at $X = 0.454$ m)
- **Vertical Tail (VT):** 10 g (at $X = 0.448$ m)
- **Battery:** 75 g (at $X = 0.010$ m)
- **Motor:** 28 g (at $X = -0.050$ m)
- **Avionics & Electronics:** 124 g (distributed around $X = 0.050$ m to 0.090 m)
- **Total System Mass:** 514 g

5 Horizontal Tail (HT) Trade Study & Selection

Table 5: Horizontal Tail Trade Study Matrix

Evaluation Criteria	Weight	0.4T	0.6T	0.8T	1.0T
Trim Angle (α at $C_m = 0$)	25%	-1	0	1	2
Static Stability (Cm_α Slope)	25%	1	1	1	2
Aerodynamic Efficiency	30%	-1	2	1	0
Structural Weight	20%	0	2	1	-1
Weighted Total Score	100%	-0.3	1.25	1.00	0.8

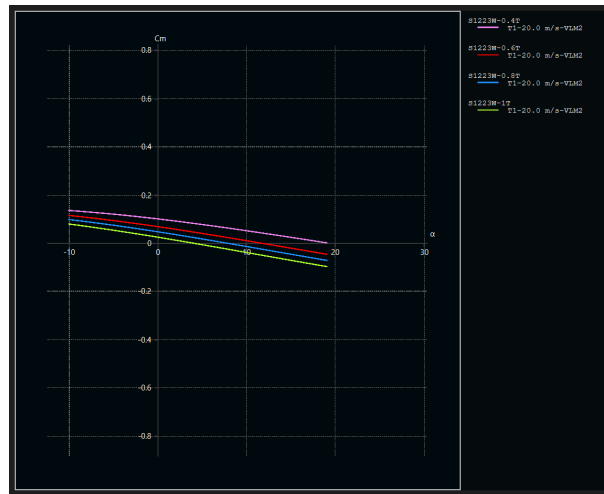


Figure 2: C_m - α graph for Horizontal Tail (0.4 , $0.6T$, $0.8T$, $1.0T$)

Selected Taper Ratio: $\lambda = 0.6$

6 Vertical Tail Design and Trade Studies

6.1 Vertical Tail Aspect Ratio (AR) Trade Study

- $AR = 1.3, 1.5, 1.7, 1.9, \text{ and } 2.1$
- $\lambda = 0.4, 0.6, 0.8, 1$
- $L_{opt} = 0.311\text{m}$

6.1.1 Vertical Tail Sizing & Numerical Derivation

- **Vertical Tail Area (S_{VT}):**

$$S_{VT} = \frac{V_{VT} \cdot S_w \cdot b_w}{L_{opt}} = \frac{0.05 \times 0.08 \text{ m}^2 \times 0.63 \text{ m}}{0.311 \text{ m}} \approx 0.0081 \text{ m}^2 \quad (6)$$

6.1.2 Trade Study Matrix

Table 6: Vertical Tail Aspect Ratio Trade Study Matrix (-2 to +2 Scale)

Evaluation Criteria	Weight	AR = 1.3	AR = 1.5	AR = 1.7	AR = 1.9	AR = 2.1
Directional Stability ($C_{n\beta}$)	40%	-1	+2	+1	+1	-1
Rolling-Yaw Coupling ($C_{l\beta}$)	30%	-1	+2	0	-1	-2
Structural Rigidity	30%	+2	+1	0	-1	-2
Weighted Total Score	100%	-0.10	+1.70	+0.30	-0.20	-1.50



Figure 3: AR = 1.3 Graph

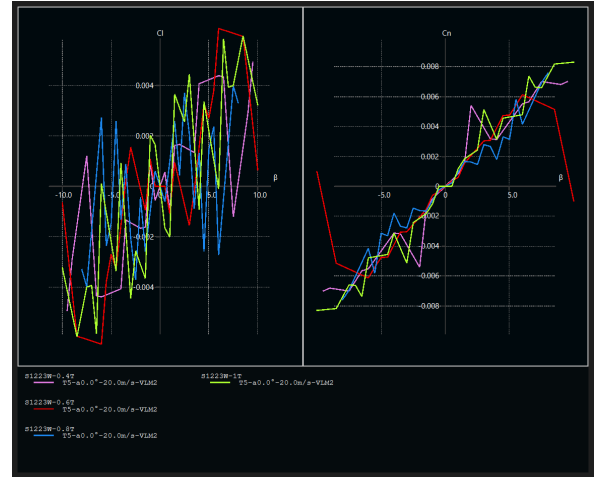


Figure 4: AR = 1.3 Graph

Selected Parameters:

- $AR = 1.5$
- $b_{VT} = 0.11 \text{ m}$
- $S_{VT} = 8.1 \times 10^{-3} \text{ m}^2$

6.2 Vertical Tail Taper Ratio Trade Study

Table 7: Vertical Tail Taper Ratio Trade Study Matrix (-2 to +2 Scale)

Evaluation Criteria	Weight	0.4T	0.6T	0.8T	1.0T
Directional Stability ($C_{n\beta}$)	40%	-1	+2	+1	0
Rolling-Yaw Coupling ($C_{l\beta}$)	30%	-1	+2	0	-1
Aerodynamic Efficiency	20%	0	+2	+1	-1
Structural Weight	10%	+1	+1	0	-1
Weighted Total Score	100%	-0.70	+2.00	+0.60	-0.50

6.2.1 Selection Conclusion

The **0.6 Taper Ratio** was used as it is the best according to the trade study

7 Modification for dynamic stability analysis

The dynamic stability analysis of the aircraft was initiated following two major geometric modifications: first, increasing the wing dihedral angle to 7° to enhance lateral stability, and second, adjusting the tail-to-wing spacing to 0.4m to improve tail effectiveness for both longitudinal and lateral stability. As a result of these modifications, the performance curves shown below were successfully obtained:

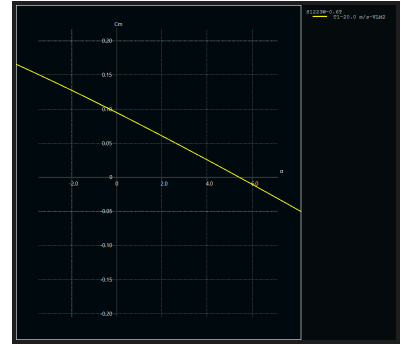
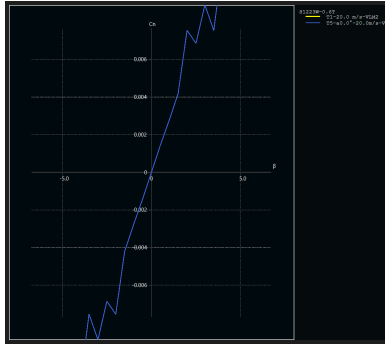
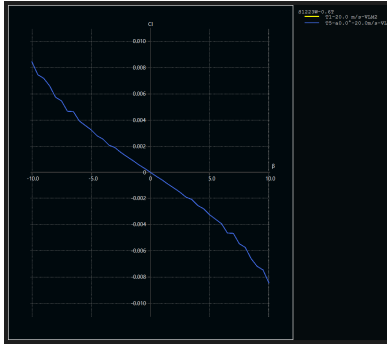


Figure 5: Rolling Moment vs. Sideslip ($C_l - \beta$)

Figure 6: Yawing Moment vs. Sideslip ($C_n - \beta$)

Figure 7: Pitching Moment vs. Angle of Attack ($C_m - \alpha$)

Figure 8: Aerodynamic stability curves after modifying the wing dihedral (7°) and the tail-to-wing distance (0.4 m).

Longitudinal Modes Graphs:

Lateral Modes Graphs:

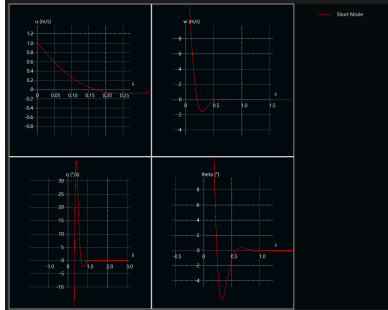


Figure 9: Short Mode Graph

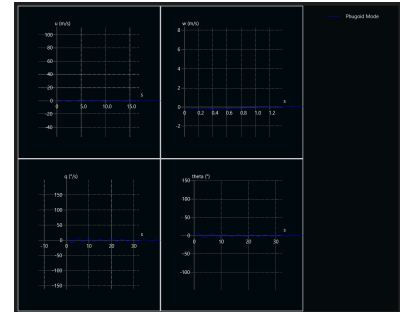
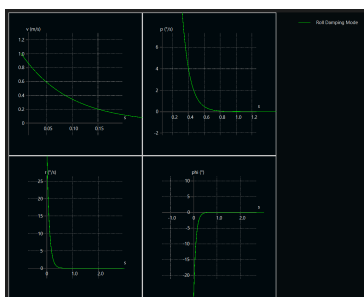
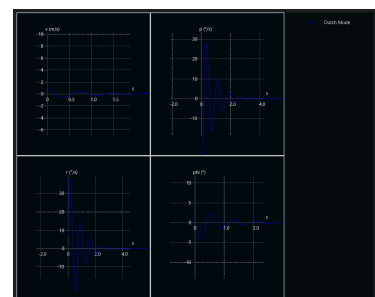


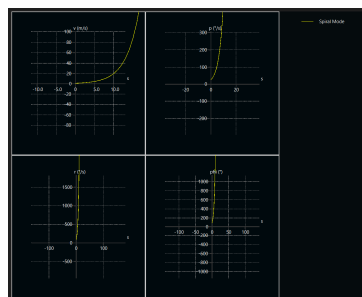
Figure 10: Phugoid Mode Graph



(a) Roll Damping Mode Graph



(b) Dutch Mode Graph



(c) Spiral Mode Graph

8 Aircraft Configuration and Reference Data for Dynamic Analysis

Table 8: Aircraft reference quantities used in AVL.

Parameter	Value
Reference area, S_{ref}	0.08001 m ²
Reference chord, c_{ref}	0.12969 m
Reference span, b_{ref}	0.63000 m
X_{ref}	0.07879 m
Y_{ref}	0.00000 m
Z_{ref}	0.00697 m
Air density, ρ	1.225 kg/m ³
Gravity, g	9.81 m/s ²
Aircraft mass, m	0.514 kg

9 Mass and Inertia Properties

The principal inertia values used in the dynamic analysis were

$$I_{xx} = 0.003232 \text{ kg m}^2, \quad (7)$$

$$I_{yy} = 0.004960 \text{ kg m}^2, \quad (8)$$

$$I_{zz} = 0.008001 \text{ kg m}^2. \quad (9)$$

The corresponding inertia tensor was

$$\mathbf{I} = \begin{bmatrix} 0.003232 & 0 & 1.89 \times 10^{-5} \\ 0 & 0.004960 & 0 \\ 1.89 \times 10^{-5} & 0 & 0.008001 \end{bmatrix} \text{ kg m}^2. \quad (10)$$

10 AVL Analysis Setup and Operating Point

Table 9: Aerodynamic operating point obtained from AVL

Quantity	Value
Velocity, V	20.0 m/s
Angle of attack, α	-8.303°
Lift coefficient, C_L	0.25723
Drag coefficient, C_D	0.01696
Pitching moment coefficient, C_m	+0.21928
C_X	-0.05393
C_Z	-0.25208
Induced drag coefficient, C_{D_i}	0.016958
Span efficiency, e	0.2679
C_L	0.25723

11 Static and Dynamic Stability Derivatives

Table 10: Static stability derivatives obtained from AVL.

Derivative	Value
C_{L_α}	4.757761
C_{L_β}	0
C_{Y_β}	-0.246263
C_{l_β}	-0.081797
C_{m_α}	-0.895599
C_{m_β}	0
C_{n_β}	0.112811

Table 11: Dynamic stability derivatives obtained from AVL.

Derivative	Value	Derivative	Value
C_{L_p}	0	C_{m_p}	0
C_{L_q}	9.007163	C_{m_q}	-16.254972
C_{L_r}	0	C_{m_r}	0
C_{Y_p}	-0.236351	C_{n_p}	0.016380
C_{Y_q}	-0.000002	C_{n_q}	-0.000002
C_{Y_r}	0.309552	C_{n_r}	-0.155319
C_{l_p}	-0.426048		
C_{l_q}	0		
C_{l_r}	0.136908		

12 Neutral Point and Static Margin

The neutral point calculated by AVL was $X_{NP} = 0.103210$ m. With $X_{CG} = 0.07880$ m and $c_{\text{ref}} = 0.12969$ m, the static margin is:

$$SM = \frac{X_{NP} - X_{CG}}{c_{\text{ref}}} = \frac{0.103210 - 0.07880}{0.12969} \approx 0.188 \implies \boxed{SM \approx 18.8\%} \quad (11)$$

13 Linearized State-Space Model and Eigenvalue Analysis

AVL generated the linearized equations of motion in state-space form: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$.

Table 12: Eigenvalues obtained from AVL Run Case 1.

Mode	Eigenvalue	Type
1	-23.800234	Real
2-3	$-1.5364187 \pm 13.028574i$	Complex pair
4-5	$-16.775574 \pm 17.971889i$	Complex pair
6	+0.096084811	Real
7-8	$+0.0033943544 \pm 1.3366623i$	Complex pair

13.1 Dynamic Mode Identification

The eigenvalue spectrum corresponds to five principal aircraft dynamic modes:

1. **Roll Subsidence:** $\lambda = -23.800234$ (Stable, time constant $\tau \approx 0.0420$ s).
2. **Dutch Roll:** $\lambda = -1.5364 \pm 13.0286i$ ($\omega_n = 13.119$ rad/s, $\zeta = 0.117$, stable but lightly damped).
3. **Short-Period Mode:** $\lambda = -16.7756 \pm 17.9719i$ ($\omega_n = 24.583$ rad/s, $\zeta = 0.682$, stable and strongly damped).
4. **Spiral Mode:** $\lambda = +0.09608$ (Unstable, doubling time $t_2 \approx 7.22$ s).
5. **Phugoid:** $\lambda = +0.00339 \pm 1.3367i$ ($\zeta = -0.00254$, very slightly unstable, doubling time ≈ 204.2 s).

Table 13: Summary of the aircraft dynamic modes.

Mode	σ	ω	ω_n	ζ	Stability
Roll subsidence	-23.800	0	23.800	1.000	Stable
Dutch roll	-1.536	± 13.029	13.119	0.117	Stable
Short period	-16.776	± 17.972	24.583	0.682	Stable
Spiral	+0.0961	0	0.0961	—	Unstable
Phugoid	+0.00339	± 1.337	1.337	-0.00254	Very slightly unstable